

Enhancing Fake Review Detection: A Hierarchical Graph Attention Network Approach Using Text and Ratings

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Abstract—The establishment of trusting connections between buyers and sellers is essential in electronic commerce (EC) since it paves the way for mutual benefit. Users can research a business's reputation before deciding to do business with them. There are some aspects of the reputation system that are unflattering to both vendors and clients. For instance, as a consequence of their dishonesty or lack of effort while posting reviews, some users may create phantom feedback, which seems as fabricated assessments meant to boost their reputation. The proposed approach incorporates stage preprocessing, feature extraction, and model training. In order to prepare the data for the SA job, the pre-processing phase incorporates initial procedures that aid in data transformation. One way to improve the model's accuracy is to use features extraction to remove irrelevant data points. We kept using the HGAN model throughout training. This state-of-the-art method outperforms SVM and LSTM with an average accuracy of 91.26%.

Keywords—Hierarchical Graph Attention Network (HGAN), Fake Review Detection, Term Frequency-Reverse Document Frequency (TF-RDF).

I. INTRODUCTION

People who are in the market to buy something often check reviews before committing to a purchase. Before buying something, many people read reviews made by other buyers. The likelihood of purchasing the product increases if the review is mostly positive. Customers are more inclined to buy other products if the review is mainly unfavorable. Companies can learn a lot about how to improve their products and services from customer reviews, which in turn affects their bottom line. Similar to how legitimate reviews can bring in money, scammers have tried to game the system by writing evaluations that aren't true and giving an unfair assessment of a service or product. Opinion spamming is what this type of individual does; they are called spammers. Regardless of how important this endeavor is, research on automatic spammer identification is still absent. Review spam detection is far more difficult than email or web-based spam detection. How simple it is for spammers to adopt phony identities is the most crucial element[1]. The difficulty in manually identifying and categorizing reviews as either true or false makes it difficult

to acquire reliable data (gold standard) associated with reviews that are either fraudulent or real, making it difficult to build a model.

Because of its meteoric rise in popularity and size in recent years, the Internet has had and will continue to have far-reaching effects on people's daily lives. Reviews, social media, and product research take up a significant portion of the typical consumer's time online[2]. Online evaluations on a wide range of topics can be crafted using an individual's expertise as well as the feedback of others who have perused their work. A wide range of products and services may get either positive or negative reviews from these reviewers. Providers can see an increase to their bottom line when there are many positive reviews, as it shows that buyers are interested in buying the goods. On the other hand, companies could wind up losing money because of negative reviews[3]. Anyone can publish reviews, thus there's no filter to prevent people from giving companies and products unfair praise or criticism. As a result, customers can be aided in their quest for truth by verifying the veracity of reviews and opinions posted online. Since spammers attempt to evade the spam filter in real-world data, the chronological order of the reviews is crucial. However, most of the current algorithms for content-based false review identification disregard this fact.

Due to the time and money it saves, online shopping has lately experienced a boom in popularity and is becoming more and more crucial to people's lives. People seem to rely heavily on this kind of online shopping. Textual and behavioral characteristics are the two main areas of research when it comes to detecting bogus reviews[4]. Review material analysis and characterization is what we mean when we speak about text characters. Supervised ML was used. The supervised classification algorithms used include SVM[5], Naïve-Bayes, KNN, Decision trees and K-Star. Classifiers including as Maximum Entropy Classifier and Random Forest were used alongside methods like SVM, Decision Tree, and Naïve-Bayes[6]. Average GRNNs, RNNs, CNNs, Bi-directional Average GRNNs, GRNNs to classify datasets. The ability of machine learning to automatically train and enhance algorithms through pattern analysis of data and subsequent

application to fresh data is one of its advantages. The semi-supervised methods offer a compromise between the two extremes of supervised and unsupervised machine learning [7]. Pattern recognition and statistical calculations are the most common uses of the K-Nearest Neighbor (KNN) approach[8]. Furthermore, logistic regression constructs a hyperplane connecting two separate data sources by means of the logistic or log function. In order to fight misinformation, a group of researchers from around the world has come together.

An AI model was created using NLP and conventional ML to categorize content as false news [9]. Articles found online and the Kaggle dataset were used to run the tests. When compared to other ML and DL models, CNN with global max pooling and random forest both showed much better accuracy. Machine learning methods were found to be approximately 6.1% less accurate when contrasted with neural network models[10]. A multilayer perceptron (MLP) neural network and an emotional and novelty-based news representation were employed by the authors of the study to counteract the propagation of false news[11]. Using four datasets to implement this method yielded statistically significant improvements in accuracy[12]. To fill this need, our research thoroughly compares and contrasts a number of machine learning techniques. Using three different datasets, these methods incorporate both traditional and cutting-edge deep learning models, such as decision trees (DTs), support vector machines (SVMs), CNNs, k-NNs, bi-LSTMs, long short-term memories (LSTMs), HANs, Conv-HANs and conv-LSTMs[13]. To our knowledge, this is the largest dataset ever used for research on the identification of fake news. Moreover, we looked at a plethora of pre-trained models in our comparative analysis, including RoBERTa, BERT, ELECTRA, Distil-BERT, ELMo[14]. To the best of our knowledge, no previous study has compared the performance of other machine learning models on a false news detection task using such complex pre-trained models.

By choosing to aggregate data on different relation types, the HGAN approach augments each company's representations with additional information. In instance, the features needed to initialize the node representations are extracted using a feature extraction module. As a relational modeling module, HGAN uses initialized node representations. After that, node representations with the extra information are sent into a task-specific layer. One typical use of our approach, which is similar to the graph classification challenge, is to forecast the movements of market indexes and individual fake reviews. HGAN outperformed every prior method because it can automatically select data.

II. PROPOSED SYSTEM

The success or failure of an online shop is often determined by reviews published by actual consumers. Online shoppers often peruse customer evaluations before making a final decision. Companies can't afford to have negative, untrustworthy evaluations floating over the web since it would hurt their business. A few businesses have gone to the extreme of paying spammers to disseminate fake evaluations. These phony reviews prey on consumers' decision-making processes when they buy. Consequently, techniques for detecting biased opinions have been extensively studied in the past twelve years.

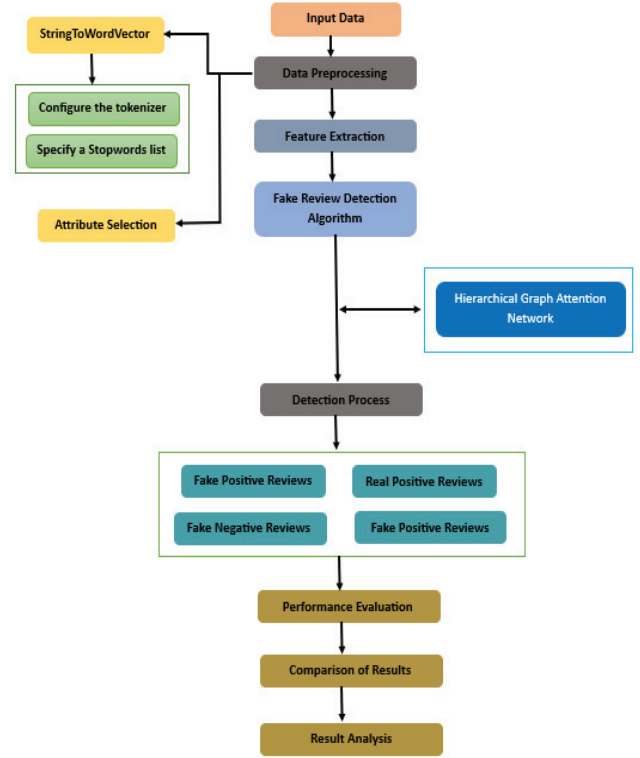


Fig. 1. Steps and Techniques used in Fake Review Detection

The objective is to achieve this by text classification analysis of a review dataset[22]. Utilizing the techniques outlined below, we follow certain SA-related steps in the suggested methodology, as illustrated in Figure 1.

A. Data Preprocessing:

The pre-processing phase includes early operations that help change the data so it is ready for the SA job. Data preparation is an essential component of many supervised learning tasks. To demonstrate its effect on classification models, our proposed approach divides data preprocessing as follows:

1) String To Word Vector:

Using the String To Word Vector filter, the principal text analysis tool in WEKA, to transform our input is the first step in preparing it for learning. This filter allows you to configure each of the numerous steps of the term extraction process. It can anticipate seeing something along these lines:

a) Configure the tokenizer:

Classification of the provided document is predicated on feature extraction, a machine learning technique that converts raw text into a set of descriptive attributes[23].

b) Specify a Stopwords List:

It is necessary to have a stop word list in order to exclude any unwanted keywords before training the classifier. The most common stop words in the English language are "the," "a," "I," "of," "you," "it," and." Common examples of such frequently used phrases are those that annoy our classifier rather than facilitate better labeling. This proposed using a database of 620 English stop words. When sorting the reviews, we can use stop word elimination to cut down on memory usage.

2) Attribute Selection:

Eliminating attributes that fail to provide a sufficient description of the data will improve classification accuracy. Reason being, not all traits are relevant for categorization, and some irrelevant ones can even make algorithms underperform. Attribute selection is the initial step in getting ready to train a classifier. Unlike unsupervised learning, which relies on data without a goal attribute, supervised learning makes use of them to direct the selection of attributes.

B. Feature Extraction:

Improving a system for pattern recognition or machine learning can be achieved through feature extraction. Feature extraction is all about stripping the data down to its essential components. The loader now has access to more pertinent information. Using features extraction to weed out superfluous data points is one strategy to boost the model's accuracy.

1) TF-RDF:

Transforming text into a comprehensible numerical representation is the primary goal of the TF-RDF algorithm. Which is used to alter prediction-related machine algorithms. The evaluation U is given by the notation $(A1, A2, A3, \dots, As)$, where Ak is the weight of each feature. The following terms are defined in relation to attribute weighting techniques:

a) Term Frequency:

Each feature's TF has been determined by keeping track of how often it appears in the review. An evaluation utilizing this method places an emphasis on the significance of a feature. What follows is the formula for TF:

$$Tf(P, U) = z(P, U) \quad (1)$$

in which Y = dataset, P = term, and U = text review

b) Inverse Document Frequency (IDF):

For evaluating the significance of features across several evaluations, IDF is one of the most used feature weighting methods, in contrast to TF, which relies on a single review to make its determination. The IDF is predicated on the idea that a feature must be easily distinguishable from other traits if it appears in few occurrences. The provided equation is used to calculate the IDF. A single sample establishes the frequency of negative evaluations[24]. This is the formula that can be used to calculate IDF:

$$(p, U) = \log \frac{S}{|\{g \in G: p \in g\}|} \quad (2)$$

S is the total number of text reviews included in the dataset.

c) TF-RDF:

Intuitive compares the feature's frequency in the training set against its frequency in a particular score. This computation establishes if an attribute is suitable for a given assessment. As an example, consider the following equation, which provides the TF-IDF formula:

$$TF - IDF(P, U, G) = tf(P, U) \times idf(P, G) \quad (3)$$

Both the TF-IDF and the BoW are feature extraction methods that can be compared and contrasted in this scenario. In contrast to TF-IDF, which assigns weights to terms based on their inverse document and term frequencies, the BoW

displays the frequency with which a word or group of words appears in a text.

C. Model Training:

1) Hierarchical Graph Attention Network:

An f -dimensional feature vector obtained by firm k 's feature extraction module at time p can be represented as $c_k^p \in R^f$. Assuming that all the node representation vectors are calculated at time step p , we omit superscript p for simplicity. Links between several kinds of relationships can be defined. In order for a graph neural network to function, we need to be aware of the subset of nodes that are neighbors of our target node k from each relation type. The collection of nodes that are immediately next to k for relation type h is represented by $S_k^{u_h}$, and the embedding vector for relation type h is represented by $c_{u_h} \in R^g$. In this case, g belongs to the relation type embedding vector. The aim to selectively collect data from nearby nodes on various relations. The want our algorithms to filter out data that won't help us predict trends. Organizations rely on this procedure since there are several kinds of relationships and some data is unrelated to movement prediction. It is typical practice to use attention processes to assign different weight values for information selection. Using a hierarchically built attention mechanism, our Hierarchical Attention Network for Stock Prediction (HATS) selectively employs relevant information at each level. Its hierarchical attention network is an essential part of improving performance. For the first state attention layer, HATS uses a cluster of nearby nodes to select important data on the same relation type. The attention method uses the updated representation of a nearby node to decide which weights to apply. To find the state attention scores, we may create a vector where $x \in S_k^{u_h}$ is placed by connecting the relation type embedding c_{u_h} with the node representations of k and l . Here is how the state attention score is calculated, using the state vector $q_{k,l}^{u_h} \in R^{(2z+g)}$:

$$o_{k,l} = q_{k,l}^{u_h} A_n + y_n \quad (4)$$

$$\beta_{k,l}^{u_h} = \frac{\exp(o_{k,l})}{\sum_i \exp(o_{k,i})} \quad (5)$$

Using the learnable parameters $A_n \in R^{(2z+g)}$ and $y_n \in R$, the state attention scores are calculated here. To create a vector representation of relation h for business k , we combine the weighted noder representations after computing the attention weight using Equation (5). This is illustrated in Equation (6).

$$n_k^{u_h} = \sum_{l \in S_k^{u_h}} \beta_{k,l}^{u_h} c_l \quad (6)$$

The aforementioned equation can be used to find every feasible representation of each relation type. In particular, we retrieved information about specific relationships from nearby nodes. One way to look at an analysis is as a summary of relational data. The vector $n_k^{u_h}$ contains the summary data from relation h . The portrayal of industry interactions offers a broad picture of the sector for four individual companies to demonstrate the idea. Our model ought to function similarly to human investors in prioritizing trading decisions according to summary information of each relationship. The design of

the second layer of HATS is to continuously assign priority weights to information utilizing another attention mechanism [25].

For the connection attention layer to take $\tilde{q}_k^{u_h} \in R^{(2z+g)}$ as input, we join the summarized relationship information vector $n_k^{u_h}$ —which shows the company's current status—with the relation type embedding vector c_{u_h} .

$$\tilde{o}_k^{u_h} = \tilde{q}_k^{u_h} A_u + y_u \quad (7)$$

$$\tilde{\beta}_k^{u_h} = \frac{\exp(\tilde{o}_k^{u_h})}{\sum_i \exp(\tilde{o}_i^{u_h})} \quad (8)$$

The weighted vectors of each connection type are added to the learnable parameters $A_u \in R^{(2z+g)}$ and $y_u \in R$ to generate an aggregated relationship representation, as indicated in Equation (9), which is equivalent to Equation (6).

$$c_k^u = \sum_i \tilde{\beta}_k^{(u_i)} n_k^{(u_i)} \quad (9)$$

Then, the demonstration of anodes is included.

$$\bar{c}_k = c_k^u + c_k \quad (10)$$

The next two sections will explain how updated node representations can be applied to various jobs.

III. RESULT AND DISCUSSION

Online reviews are a great way to gauge public opinion on a product or service, and customers use them extensively when making purchasing decisions. Because of the sway that customer reviews and ratings can have, businesses in the retail and manufacturing industries value them highly. A concern that arises when people rely on online reviews is the possibility that dishonest people may inflate review prices to influence the worth of products and services.

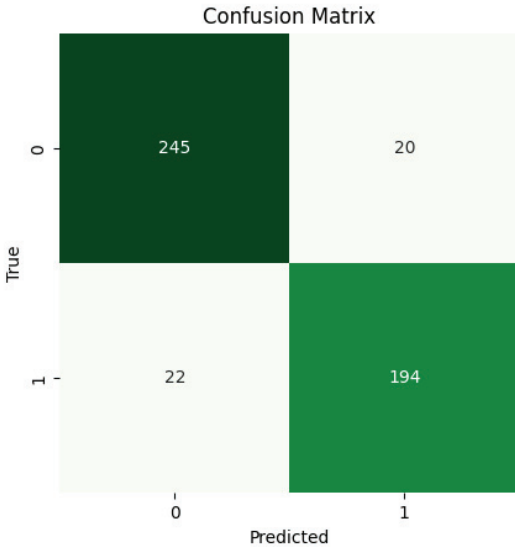


Fig. 2. Confusion Matrix for HGAN Model

The results displayed in Figure 2 The HGAN model successfully identified 245 false reviews and 194 genuine

reviews out of 481 reviews used for training. There was an error in the classification of 22 genuine reviews and 20 false reviews.

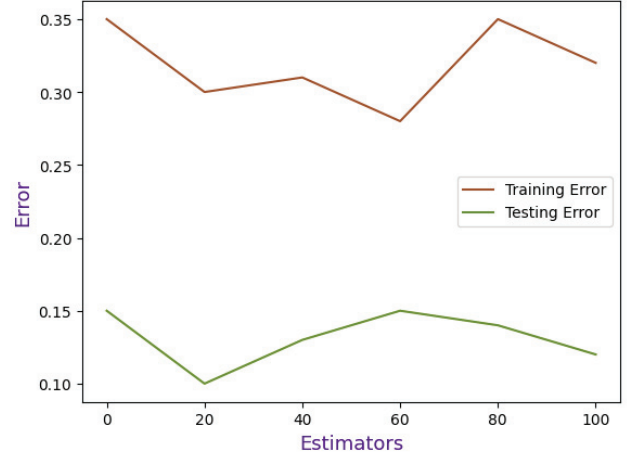


Fig. 3. Errors in Training and Testing for Various Estimator Numbers

It seems that the number of Estimators (which might mean iterations or models in an ensemble, like an HGAN) is a function of both the Training Error and the Testing Error in the given figure 3.

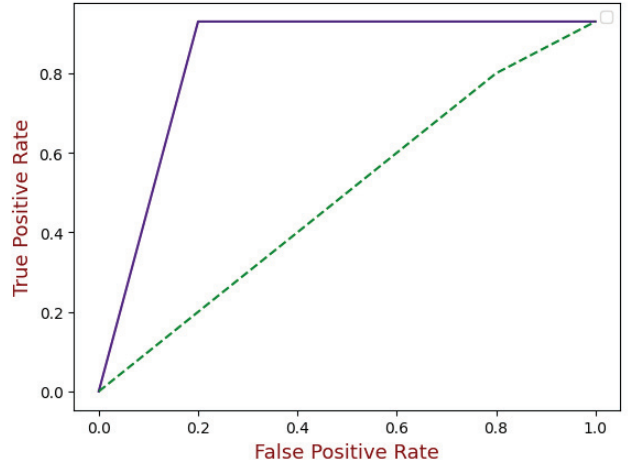


Fig. 4. ROC Curve of HGAN Models

The receiver operating characteristic curve (ROC) is a useful graphical tool for assessing the accuracy of a binary classification model, as shown in figure 4. At different categorization criteria, it shows the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR).

TABLE I. PERFORMANCE COMPARISON(%)

Models	Sensitivity	Acc	F1	Precision	Recall
DNN	0.9107	0.8104	0.7873	0.8019	0.7906
KNN	0.9182	0.8356	0.8251	0.8092	0.8164
LightBGM	0.9257	0.8532	0.8423	0.8230	0.8350
SVM	0.9356	0.8654	0.8543	0.8309	0.8433
LSTM	0.9405	0.8935	0.8828	0.8634	0.8702
HGAN	0.9452	0.9126	0.9053	0.8845	0.8950

All of the algorithms are compared in Table 1 according to the evaluation parameters. Table 1 displays the findings for each classifier used in this proposed, including F1-score, sensitivity, precision, accuracy, and recall.

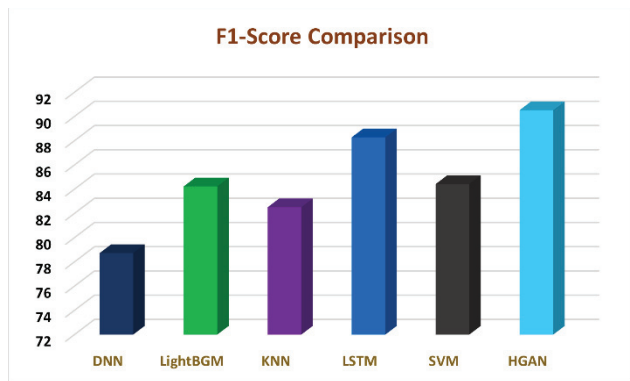


Fig. 5. F1-Score Comparison of Different Model

Figure 5 displays the results of a graphical comparison of all DL algorithms' F1-Score evaluation parameters.

IV. CONCLUSION

Internet shopping and e-commerce have experienced unprecedented popularity due to the ongoing COVID-19 pandemic. Customers can browse products and place orders from the convenience of their own homes using a computer or mobile device. Sellers' photos and descriptions are more trusted than buyers' own inspections when making purchases. Reviews provided by real customers are crucial since they assist other shoppers make an informed decision. However, reviews can be skewed to paint an unfair or misleading image of a product. Initial techniques that aid in data transformation are incorporated into the pre-processing phase to prepare the data for the SA task. Using features extraction to weed out superfluous data points is one strategy to boost the model's accuracy. It is enough to use the HGAN for training the model. The approach outperforms SVM and LSTM on average, with an accuracy of 91.26 percent.

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